

Standalone RCA progress since July 2026

Project 19 | Progress through 22 September 2026

Since the July presentation, we have developed the standalone root cause analysis system into an iterative investigation prototype, run Qwen development experiments, diagnosed failures in both execution and reasoning, and implemented changes to evidence handling and revision. We have also prepared two staged evidence packages for the next controlled evaluation.

The main progress is a working, inspectable investigation process and a clearer way to test it. We have not yet demonstrated reliable autonomous RCA or a controlled improvement in causal accuracy. This report summarizes the work so it can be used to prepare the next project presentation.

Where we were in July

The July presentation described an RCA pipeline starting from sparse incident records. It organized the investigation around a neutral narrative, entity identification, document requests, a timeline, candidate causes, pruning, and corrective actions. A facilitator review step kept a person involved before finalization. The presentation also showed an early RCA interface and causal map.

The limitations identified in July were unreliable expert knowledge, limited context retention, and attempts to perform RCA in one pass. The stated next steps were validation against actual incident RCA and testing open-source model viability. The work since then has concentrated on those problems. [1]

Main work completed

- **Iterative investigation prototype.** Built workflows that preserve the current explanation, source references, open questions and successive investigation versions.
- **Model experiments.** Ran Qwen on development incidents, including the R3 beam case and the instrument-air interruption case, and retained outputs for review.
- **System diagnosis and redesign.** Traced failures to evidence delivery, growing model context, question handling and incomplete application of corrections, alongside errors in model interpretation.
- **Implementation and test infrastructure.** Added more explicit evidence, answer-delivery, revision and review records, with exportable experiment traces.
- **Staged evaluation material.** Prepared two historically anchored synthetic investigation packages with three cumulative evidence releases. [2–5]

How the investigation now works

The prototype treats RCA as an investigation that develops as evidence arrives. A first version may legitimately contain provisional explanations and unanswered questions. Later versions should change those explanations only when the new evidence supports a change.

Incident understanding and specialist investigation

The user provides an incident description and available documents. Default references, incident starter records and later answer attachments remain distinct. The system identifies relevant entities and operating context, forms preliminary explanations, and selects relevant specialist perspectives. Tags such as equipment, stored energy or isolation direct attention; they do not establish a cause.

Specialists investigate concrete uncertainties and propose questions. The question broker combines overlapping requests while retaining their purposes and owners. Answer fetching searches released documents and available knowledge within that model's investigation. Answers retain source references, partial coverage and unresolved conflicts. Repeating one source across agents does not create independent corroboration.

Causal analysis and revision

The causal-analysis process considers how the incident could have occurred, what conditions an explanation requires, and what evidence would distinguish alternatives. Review separates three questions: whether a statement is supported, whether it has the proposed causal role, and whether a connection between statements is justified. A supported gauge reading, for example, does not automatically establish a protective barrier.

The application publishes a saved version containing the working board, evidence and open questions. New records can lead to corrections, changed explanations or renewed uncertainty. Earlier versions remain available for comparison. These capabilities are implemented to varying levels of acceptance; their presence does not guarantee correct model judgments. [2, 4]

What each development generation contributed

- **Try 4.** Explored a specialist-led evidence and question pipeline that generated causal boards. Saved runs revealed incomplete causal integration and excessive growth in shared context and candidate-link work.
- **Try 5.** Introduced durable tasks, source-bound evidence and more local handling of failures. Recovery became more inspectable, but substantial work could still occur before a useful investigation finished.
- **Try 6.** Established an initial investigation position earlier, used targeted consultations and published partial V1 and V2 investigations. Review showed that new evidence did not consistently revise explanations or correct current board wording.
- **Try 7.** Implemented core changes to question coverage, answer delivery, sourced revisions and separate fact, role and connection checks. Full-plan acceptance and controlled model validation remain incomplete. [2, 4]

Try 4 through Try 7 are application generations. V1, V2 and V3 are saved investigation versions within a model's run. The development runs used changing configurations and are not a controlled ranking of application generations.

What the experiments showed

Instrument air investigation with Try 6

We preserved two partial Qwen investigations for IA-16036, an instrument-air interruption during maintenance preparation. V2 received routing and valve records plus one released answer. The comparison showed more observations and a larger board, but no verified causal connections. [6]

Recorded measure	V1	V2
Uploaded documents excluding the incident description	10	12
Released answers	0	1
Board nodes including the focal event	5	9
Active connections	4	6
Verified connections	0	0
Published status	Partial	Partial

The important finding was not that V1 remained incomplete. The available records already supported facts that the system sometimes misrepresented, and later evidence did not consistently narrow its explanations.

- **Chronology.** The board retained opposing “preceded” connections between pressure loss and recovery, even though recovery should follow the interruption.
- **Factual correction.** A pressure timestamp error remained in the investigation. Review feedback did not reliably propagate into the wording of the current board.
- **Causal interpretation.** The model treated a compressor running indication as a possible barrier without establishing a protective function.
- **Evidence use.** Some requests still asked for information already supplied, while new routing and valve observations did not produce the expected mechanism revision.

Latest Try 7 live Qwen smoke run

The unscored pneumatic-carriage smoke run reached a preserved partial V1 on 20 September. It completed 21 scheduled tasks and published five selected observations plus the focal event, with zero causal links. The snapshot retained 20 rejected output items and one pending answer-delivery receipt.

The run exposed additional question-identity, assignment and revision-handling defects. We implemented fixes and regression checks. However, configuration changes and a recorded checkpoint repair occurred during the run, and the post-run fixes still require a fresh run with a frozen configuration. Its assessment remains execution-inconclusive, not an RCA performance score. [4]

Changes made after reviewing the runs

The reviews showed that the application must reliably preserve, deliver and update evidence before a result can fairly assess model reasoning. We retained the investigation workflow and changed how its records and judgments move through the system. The Try 7 implementation receipt documents the following work. [3, 4]

- **Preserved baselines and isolated development.** Created Try 7 with separate local storage and execution identity. The implementation record reports unchanged fingerprints for 268 baseline source, prompt, configuration and test files in Try 6.
- **More explicit explanations.** Extended explanation records with assumptions, required conditions, alternatives and observations that could distinguish them. This makes an explanation easier to inspect and challenge.
- **Question coverage and delivery.** Tracked answers by subpart, retained different subscriber purposes, and recorded delivery to relevant specialists and explanations. Reviewer-generated questions gained a visible answer path.
- **Applied revisions.** Added a focused revision stage with source references, expected record revisions and retained history. A correction must pass review; proposing it does not make it supported.
- **Separate review judgments.** Separated factual support, causal-role validity and connection validity. Added structural checks for issues such as inconsistent temporal direction and stale premises.
- **Controlled releases and exports.** Added release receipts tied to a model track and parent version, along with experiment exports containing saved inputs, outputs, snapshots, task records and usage. These exports are unscored records.
- **Clearer execution status.** Kept provider failures, application limitations and evidence uncertainty distinguishable. A completed software workflow does not automatically establish a correct root cause.

Recorded software verification

The implementation receipt reports 97 deterministic checks: 20 engine tests, 24 Try 7 tests, 37 regression tests and 16 capacity tests. TypeScript and application build checks also passed. Browser and HTTP checks exercised the local interface and export routes.

These results support the implemented software contracts. They do not establish industrial RCA accuracy, remove every possible application defect, or replace a controlled live-model pilot. The smoke run found issues that scripted checks had missed, which is why the next frozen-configuration run remains necessary.

What we learned

Larger context limits, additional model calls and larger causal boards did not reliably solve the investigation problem. The system needs the right original evidence for each judgment, explicit uncertainty, and a revision path that changes the saved explanation when a correction is accepted. Execution reliability and causal quality must be evaluated separately.

Evidence packages and the next milestone

Historical incident rows anchor the event but often omit the records needed to justify a detailed causal reconstruction. We prepared two staged, historically anchored synthetic investigation packages on 22 September. They are constructed test material, not recovered company records. [5]

- **R3 carrier-beam fall.** The package includes staged evidence for a beam-fall scenario. Its load-path and circuit reconstruction requires independent technical review before mechanical conclusions support scored claims.
- **Instrument-air interruption.** The package combines configuration, pressure observations and operating chronology across separate records. It is prepared for a pilot, but has not received independent expert validation.

Both packages use three cumulative releases. B0 provides the initial account and starter evidence for V1. B1 supplies information intended to distinguish explanations for V2. B2 adds deeper context or corrective evidence for V3. Private scenario records and expected changes are separated from investigator-facing material. These packages derive from previously used development incidents, so they are not independent held-out test cases. The package record reports no completed RCA runs on these new packages.

Next steps

The next milestone is one reproducible V1–V3 investigation. We need to freeze the model, prompts and evidence schedule, preserve each version, and inspect whether new evidence changes the explanation for justified reasons. A simpler baseline should receive the same evidence for comparison. Assessment should cover factual fidelity, supported causal mechanisms, uncertainty, useful questions and revision quality, with execution failures and resource use reported separately. Technical review should assess scenario realism and causal interpretation before broader evaluation. [7]

Project summary

Since July, we have built and exercised a more structured RCA investigation system. We can preserve successive versions, inspect evidence and questions, and trace where an investigation fails. Development experiments identified specific weaknesses, which informed the latest implementation changes. We now have staged evidence packages for a controlled pilot. The remaining research question is whether these changes improve evidence-supported causal explanations and their revision as new information arrives.

Source records

- [1] S&T PTC July 2026 Presentation, slides 7–13.
- [2] RCA Research Overview and Plan, 21 September 2026, sections 5 and 8.
- [3] RCA System Review, 18 September 2026, sections 2 and 3.
- [4] Try 7 Executive Plan, section 10 implementation receipt and observed smoke outcome.
- [5] Synthetic Evidence Packages 2026-09-22, operator overview and case status.
- [6] Try 6 V1 V2 Evidence Review, 20 September 2026, sections 1–3.
- [7] RCA Implementation and Paper Checklist, 20 September 2026, sections C–E.